GROUP PRESENTATION

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INTRODUCTION

- The Fibonacci series is a sequence of numbers where each number is found by adding the two numbers before it.
- The series starts with 0 and 1 and continues As follows,,,0,1,1,2,3,5,8,13,21,34 and so on
- In this case study wee look at ways of generating Fibonacci numbers using recursion , iteration and dynamic programming
- We also apply the series to a customer Loyalty reward system and compare the methods

OBJECTIVES

- TO IMPLEMENT FIBONACCI USING RECURSION
- TO GENERATE FIBONACCI NUMBERS USING ARRAYS AND LOOPS AND ARRAYS
- TO UNDERSTAND DYNAMIC PROGRAMMING USING MEMOZATION
- TO APPLY THE SERRIES TO A COSTUMER LAYALTY REWARD SYSTEM
- TO COMPARE PERFORMANCE FOR N EQUALS TO 30 AND RECOMMEND A SUITABLE METHOD

RECURSIVE APPROACH

- Recursion is when a problem calls itself to solve a smaller part of a problem
- In Fibonacci , it calls itself to calculate the two previous numbers until it reaches 0 or 1

JAVA CODE

COMPLEXITY

- THE SIMPLE RECURSIVE METHOD TAKES EXPONENTIAL TIME ,,,OFTEN DESCRIBED USING THE UPPER BOUND 0[2^n], because it repeats many calculations
- Its space complexity is 0 (n) because of the recursive call stack
- It is easy to understand but becomes slow for larger numbers